PHASE 2

1.Wazuh

Wazuh is an open-source:

- SIEM (Security Information and Event Management)
- XDR (Extended Detection and Response)

It helps organizations:

- Collect logs
- Detect threats
- Monitor endpoints

- Generate alerts
- Respond to incidents

Windows/Linux Endpoint
↓
Wazuh **Agent**
↓
Wazuh **Manager**
↓
OpenSearch **Indexer**
↓
Wazuh **Dashboard**
↓
SOC Analyst

1.Wazuh Agent

The agent is installed on endpoints.

Examples:

- Windows PC
- Linux server
- macOS system

What does it do?

It collects:

- Login events
- Process creation events
- File changes
- Security logs

Example:

```
User: diya
Failed Login
Time: 10:30 AM
```

The agent collects this log and sends it to the manager.

2.Wazuh Manager

Think of this as the **brain**.

Responsibilities

- Receives logs from agents
- Analyzes events
- Applies rules
- Generates alerts

Example:

```
50 failed logins detected
```

The manager checks its rules.

Rule:

```
More than 10 failed logins
```

Result:

```
Brute Force Alert Generated
```

3.Wazuh Indexer

Stores all logs and alerts for searching and investigation.

Built on:

OpenSearch

Responsibilities

- Store data
- Index data
- Enable fast searching

Without the Indexer:

```
No history
No searching
No investigations
```

4.Wazuh Dashboard

This is where analysts work.

Functions

- View alerts
- Search logs
- Investigate incidents
- Monitor agents
- Create reports

Typical analyst workflow:

```
Alert Appears
↓
Open Alert
↓
Investigate
↓
Classify
↓
Escalate or Close
```

Example: How an Alert is Generated

Step 1

Attacker tries many passwords:

```
admin
admin123
password
root
```

Step 2

Agent collects logs.

Step 3

Manager analyzes logs.

Step 4

Rule matches.

```
Failed logins > 5
```

Step 5

Alert created.

```
Possible Brute Force Attack
Severity: High
```

Step 6

Alert appears in Dashboard.

SOC analyst investigates.

Wazuh Components Summary

Component	Purpose
Agent	Collects logs from endpoints
Manager	Processes logs and generates alerts
Indexer	Stores and indexes data
Dashboard	Used by analysts for monitoring and investigation

2.Rule Levels

Level	Severity
0–3	Informational
4–6	Low
7–9	Medium
10–12	High
13–15	Critical

Rule Structure

```
Rule Components

- Rule ID → Unique identifier
- Rule Level → Severity (0–15)
- Description → What the rule detects
- Group → Alert category
```

Example:

```
<rule id="5710"level="5">
```

Meaning:

```
Rule ID = 5710
Rule Level = 5 (Low Severity)
```

3.File Integrity Monitoring (FIM)

Purpose:

Detect file changes.

Example:

```
C:\Windows\System32\hosts modified
```

Alert generated.

4.Active Response

Purpose:

Automatic response.

Example:

```
Brute Force Detected
↓
Block IP Automatically
```